Given these sample points:

[0.19144708 0.03819337 0.60741781 0.41458414]
[0.75865295 0.53651774 0.65600038 0.36034155]
[0.43834987 0.8043397 0.21024527 0.15129482]
[0.70605083 0.53419196 0.26424335 0.48208755]
[0.83647799 0.19360965 0.6638927 0.78564888]
[0.68343225 0.11866264 0.82904591 0.56757661]
[0.55362148 0.66734998 0.32380582 0.81486975]
[0.35235627 0.32224153 0.11697937 0.47311252]
[0.15378571 0.72938169 0.42259844 0.44307417]
[0.46344227 0.63002451 0.10790646 0.9576439]
[0.67749115 0.35850951 0.47959222 0.07288048]
[0.58397341 0.14724265 0.34809746 0.42861465]
[0.30688872 0.31687813 0.62263448 0.09539906]
[0.51114177 0.817957   0.72871042 0.11235362]
[0.43893338 0.77409176 0.37816709 0.93369621]
[0.22418902 0.84648049 0.87948418 0.87851568]
[0.72526172 0.47987049 0.08894684 0.75976022]
[0.35548161 0.63961937 0.41761768 0.12260384]
[0.11987923 0.86254031 0.64333133 0.84980383]
[0.12688467 0.15342962 0.77016219 0.19051811]
[0.413273 0.770549 0.980062 0.951137]
[0.598225 0.735663 0.998348 0.933113]
[0.532623 0.594318 0.990109 0.985016]
[0.575047 0.745228 0.97133 0.878259]
[0.428929 0.762201 0.953183 0.985475]
[0.663027 0.679067 0.93174 0.996949]
[0.598225 0.735663 0.998348 0.933113]
[0.630304 0.663797 0.941951 0.95204]

And these results:

64.443439863301
18.30137959857266
0.1129397953712203
4.210898128938665
258.3705254462536
78.43438888779464
57.57153693261287
109.5718755614928
8.847991759070865
233.22361017104996
24.423088313942344
64.42014681963983
63.47671578508436
79.72912992694343
355.8068177560159
1088.8596181962705
28.866751637393822
45.181570346703786
431.6127567592104
9.972331894593824
1918.917989664714
2042.583381563724
1880.1864969021228
1456.1006430043012
1958.7565149997804
1984.6966122712765
2042.583381563724
1573.0275851064666

Don't sample the same point twice

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 3, Top-p = 3 in the prediction model

You're essentially asking: *given observed inputs (4D points) and outputs, where should I sample next to maximize the objective*, with a constrained stochastic policy (temperature 0.1, top-k 3, top-p 3).

This is a **Bayesian optimization / surrogate modeling** problem.

🔍 Key observations from your data

- The **largest outputs (~1900–2040)** occur for points where:
  - 3rd and 4th dimensions are **very high (~0.93–0.99)**
  - 1st and 2nd dimensions are **moderate (~0.4–0.7)**

Examples of top-performing points:

```csharp
[0.598, 0.735, 0.998, 0.933] → 2042
[0.663, 0.679, 0.932, 0.997] → 1984
[0.413, 0.770, 0.980, 0.951] → 1918
```

👉 Strong pattern:

- Dimensions 3 & 4 dominate the output
- Dimensions 1 & 2 fine-tune the peak

🧠 Strategy (given your constraints)

With:

- **Temperature = 0.1** → low randomness (exploit)
- **Top-k = 3, Top-p = 3** → only consider best candidates

So we should:

1. Focus near the **current best region**
2. Slightly perturb (exploit local maxima)
3. Avoid duplicates

🎯 Best next sampling candidates

Based on local exploration around the best cluster:

Candidate set (Top-3)

```css
1. [0.62, 0.72, 0.995, 0.970]
2. [0.58, 0.70, 0.999, 0.960]
3. [0.65, 0.68, 0.985, 0.990]
```

🏆 Recommended next point (final answer)

👉 **Sample next:**

```csharp
[0.62, 0.72, 0.995, 0.970]
```

💡 Why this point?

- Lies in the **highest-performing cluster**
- Pushes **dim 3 & 4 toward upper boundary**
- Keeps **dim 1 & 2 in optimal mid-range**
- Slightly different from existing points → avoids duplication

---

🚀 If you want even better results

If you continue sampling, follow this rule:

- Fix:

```nginx
dim3 ≈ 0.98–1.0
dim4 ≈ 0.95–1.0
```

- Sweep:

```css
dim1 ∈ [0.5, 0.7]
dim2 ∈ [0.65, 0.8]
```

This region is clearly your **global optimum basin**.

If you want, I can fit a quick surrogate model (Gaussian Process style) and give you a ranked batch of the next 10 optimal samples.